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CMP 262

12 May 2026

Recalls Analysis - Project Summary

Tools Used

To complete this project, I used Jupyter Notebook with Python. The packages I used in Python included Pandas for data cleaning and Matplotlib for data visualizations. Excel was also used to do a preliminary analysis.

Questions Asked

The questions asked in this analysis what is the most apparent reason for recalls, what is the average amount of vehicles affected per recall, which auto manufacturer is the least reliable (most # of recalls), and has the amount of recalls gone up or down over the past 5 years.

Insights Discovered

Through analyzing the data, it is seen that the most unreliable manufacturers are domestic (American), and the parts with the most recalls relate to the braking systems and tires of cars. While there has been some variability in the amount of recalls throughout the past five years, it has generally stayed in the 1000-1100 range. Finally, the average amount of vehicles affected per recall are ~45k, however the median amount of vehicles affected per recall is 679, showing how volatile recalls can be.

Recommendations

If this information was to be presented to a client, I would recommend to quality check each batch of tires recieved from third party sources, or even attempt themselves to create their own tires. The same applies for braking systems.

Future

For the future, I would like to incorporate and use more datasets, including if these recalls had any direct involvement in accidents, as well as how the amount of recalls a company has put out affects their sales.